PACKETS FROM CONTROLLER BOARD

Packets from controller board to pi are sent MSB first.
Total packet length is 3 bytes, 24bits.
MSB is the command opcode, the next 2 are data.

Any of these commands are possible from the controller board.

```
SWStateCMD = 0x01
OpticalStateCMD = 0x02
TOF1DistCMD = 0x03
TOF2DistCMD = 0x04
MotorStatusCMD = 0x05
LeftMotorCMD = 0x07
RightMotorCMD = 0x08
FWVersionCMD = 0x09
TemperatureCMD = 0x0B
LeftMotorCurrentCMD = 0x0C
RightMotorCurrentCMD = 0x0D
```

COMMAND								DATA															
b23	b22	b21	b20	b19	b18	b17	b16	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Limit Sensor/Door Switch CMD 0x01 Switch Closed bit=0, Switch Open bit=1, x = don't care

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
x	x	x	x	x	x	x	x	x	DOOR_SW	SW_1A	SW_1B	SW_1C	SW_2A	SW_2B	SW_2C

Optical Sensor CMD 0x02 Optical Beam Broken bit=0, Beam Not Broken bit=1, x = don't care

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
x	x	x	x	x	x	x	x	x	x	x	x	OPT_1B	OPT_1A	OPT_2B	OPT_2A

TOF Sensor 1 CMD 0x03 Distance in millimeters, 0 to 65535mm, upper limit TBD

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm

TOF Sensor 2 CMD 0x04 Distance in millimeters, 0 to 65535mm, upper limit TBD

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm

Left Motor Stop/Run bit0, Right Motor Stop/Run bit4, Stop=0, Run=1
Motor Status CMD 0x05 Left Motor Stuck bit1, Right Motor Stuck bit5, Normal=0, Stuck=1 x = don't care

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
x	x	x	x	x	x	x	x	x	x	R_STUCK	R_STOP	x	x	L_STUCK	L_STOP

Direction bit7, Forward bit=0, Reverse bit=1

Left Motor CMD 0x07 Speed bit0-6, 0 to 100, 0 is Stop, 100 is Full Speed, x = don't care

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
x	x	x	x	x	x	x	x	F/R	speed	speed	speed	speed	speed	speed	speed

Direction bit7, Forward bit=0, Reverse bit=1

Right Motor CMD 0x08 Speed bit0-6, 0 to 100, 0 is Stop, 100 is Full Speed, x = don't care

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
x	x	x	x	x	x	x	x	F/R	speed	speed	speed	speed	speed	speed	speed

Firmware Version CMD 0x09 Minor Version bit0-3, 0 to 15, Major Version, bit4-7, 0 to 15, x = don't care

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
x	x	x	x	x	x	x	x	Major	Major	Major	Major	Minor	Minor	Minor	Minor

Temperature CMD 0x0B Temperature in Celsius bit0-7, signed byte -128 to 127, x = don't care

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
x	x	x	x	x	x	x	x	temp	temp	temp	temp	temp	temp	temp	temp

Left Motor Current CMD 0x0C Current in mA bit0-15, 0 to 65535mA, upper limit TBD

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA

Right Motor Current CMD 0x0D Current in mA bit0-15, 0 to 65535mA, upper limit TBD

DATA															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA	mA

PACKETS TO CONTROLLER BOARD

Packets from pi to controller board are sent MSB first.
Total packet length is 2 bytes, 16bits.
MSB is the command opcode, LSB is data.

Any of these commands are possible from the pi board.

```
ControllerStatusCMD = 0x06
LeftMotorCMD = 0x07
RightMotorCMD = 0x08
```

All commands from the above can be called anytime to the controller board, to retrieve the needed data.
The following command op-code can also be used with CMD 0x06, to enter bootloader state, or reboot the controller board.

```
BootloaderCMD = 0x0A
RebootCMD = 0x0E
```

The bootloader and reboot commands will not return anything.

Controller Status CMD 0x06 Command Op-Code from above.

COMMAND								DATA							
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
0	0	0	0	0	1	1	0	CMD	CMD	CMD	CMD	CMD	CMD	CMD	CMD

Direction bit7, Forward bit=0, Reverse bit=1
Left Motor CMD 0x07 Speed bit0-6, 0 to 100, 0 is Stop, 100 is Full Speed

COMMAND								DATA							
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
0	0	0	0	0	1	1	1	F/R	speed	speed	speed	speed	speed	speed	speed

Direction bit7, Forward bit=0, Reverse bit=1
Right Motor CMD 0x08 Speed bit0-6, 0 to 100, 0 is Stop, 100 is Full Speed

COMMAND								DATA							
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
0	0	0	0	1	0	0	0	F/R	speed	speed	speed	speed	speed	speed	speed

NOTES:

When '0x06 0x0A' is sent to the controller board, USB will disconnect from serial mode and connect to DFU mode.
On the pi side, need to install and run dfu-util, https://dfu-util.sourceforge.net/
The DFU command is **dfu-util -a 0 -s 0x08010000 -D /path/to/image.bin -R**